

Yixiao Wang

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Durham, NC 27705, USA

EDUCATION

- Duke University** Aug. 2024 – May. 2026
M.S. in Statistical Science
◦ GPA: 3.96/4.00
Durham, NC, United States
- University of Science and Technology of China, School of the Gifted Young** Sep. 2020 – Jul. 2024
B.S. in Mathematics
◦ Top 10% in School of the Gifted Young & Mathematics; Outstanding Graduate (2024)
◦ Thesis: *Trimmed Mean for Partially Observed Functional Data* (Advisor: [Dr. Xiaohong Lan](#))
Hefei, Anhui, China

RESEARCH INTERESTS

- Machine learning theory**, with emphasis on scalable and theoretically grounded algorithms.
- Deep learning theory**, aiming to uncover principles for efficient, interpretable, and practically deployable systems.

RESEARCH EXPERIENCE

- Duke University, Interpretable Machine Learning Lab** Jun. 2025 – Present
Supervisor: [Dr. Cynthia Rudin](#) Durham, NC, United States
 - Developing scalable algorithms for ϵ -optimal sparse regression trees with continuous features, leveraging fast rank-one Cholesky updates to enable efficient and numerically stable split evaluation.
 - Investigating theoretical guarantees and computational trade-offs of sparse lookahead strategies for interpretable regression trees.
- Duke University, Department of Computer Science** Jun. 2025 – Present
Supervisor: [Dr. Anru Zhang](#) Durham, NC, United States
 - Established the first rigorous in-context learning theory for time series forecasting, showing that Transformers are fundamentally suboptimal: proved a strictly positive excess-risk gap over linear predictors, quantified its $\mathcal{O}(1/n)$ decay rate, and demonstrated exponential error compounding under Chain-of-Thought rollout.
 - Co-first authored a paper based on this work, currently under review at *ICLR 2026*.
- Duke Center for Computational Evolutionary Intelligence (CEI)** Jan. 2025 – Present
Supervisor: [Dr. Yiran Chen](#) Durham, NC, United States
 - Developed **ZEUS**, a controllable training-free diffusion acceleration framework achieving comprehensive $2\times-4\times$ speedups while maintaining perceptual fidelity. The paper based on this work, currently under review at *ICLR 2026*.
 - Developed **SADA** (ICML 2025), a stability-guided training-free adaptive diffusion accelerator, delivering $\geq 1.8\times$ speedups across SD-2, SDXL, and Flux with LPIPS ≤ 0.10 and FID ≤ 4.5 .
- Wake Forest University, Department of Computer Science** Jan. 2025 – Present
Supervisor: [Dr. Aditya Devarakonda](#) Winston-Salem, NC, United States
 - Proposed and led the development of **Enhanced Cyclic Coordinate Descent (ECCD)**, a block coordinate descent algorithm with second-order Taylor correction for GLMs, ensuring both efficiency and stability even on large-scale datasets requiring tens of minutes to hours of fitting.
 - Achieved consistent $2-4\times$ speedups over the widely used `glmnet` package while preserving convergence guarantees and solution accuracy.
 - Co-first authored a paper based on this work, currently under review at *NeurIPS 2025*.

PUBLICATIONS AND PREPRINTS

C=CONFERENCE, S=SUBMISSION, P=PREPRINT, I=IN PREPARATION

- [C.1] Ting Jiang*, **Yixiao Wang***, Hancheng Ye*, Zishan Shao, Jingwei Sun, Jingyang Zhang, Zekai Chen, Jianyi Zhang, Yiran Chen, Hai Li. **"SADA: Stability-guided Adaptive Diffusion Acceleration,"** in *ICML*, 2025. [Code] [Page] [Slides]
- [S.1] **Yixiao Wang***, Zishan Shao*, Ting Jiang, Aditya Devarakonda. **"Enhanced Cyclic Coordinate Descent for Elastic Net GLMs,"** Manuscript submitted to *NeurIPS*, 2025.
- [P.1] Yufa Zhou*, **Yixiao Wang***, Surbhi Goel, Anru Zhang. **"Why Do Transformers Fail to Forecast Time Series In-Context?,"** *arXiv preprint*, 2025. (Submitted to *NeurIPS 2025 Workshop on What Can't Transformers Do?*)
- [I.1] **Yixiao Wang***, Ting Jiang*, Zishan Shao*, Hancheng Ye, Jingwei Sun, Mingyuan Ma, Jianyi Zhang, Yiran Chen, Hai Li, 'ZEUS: Zero-shot Efficient Unified Sparsity for Generative Models,' Manuscript in preparation (target: *ICLR 2026*). [Code] [Page];
- [S.2] Zishan Shao, **Yixiao Wang**, Qinsi Wang, Ting Jiang, Zhixu Du, Hancheng Ye, Danyang Zhuo, Yiran Chen, Hai Li. **"FlashSVD: Memory-Efficient Inference with Streaming for Low-Rank Models,"** Manuscript submitted to *AAAI*, 2026. [Code]

* denotes co-first authorship; first-/co-first-authored works listed first.

HONORS AND AWARDS

- Outstanding Graduate, University of Science and Technology of China (USTC) 2024
- China Petroleum Scholarship (3 awardees, School of the Gifted Young, USTC) 2023
- Excellent Teaching Assistant (Top 3 in the University, USTC) 2023
- Silver Prize, Outstanding Student Scholarship (Top 15%, USTC) 2022 & 2021
- “Qiang Wei Feng Gong De Yu” (Diligence and Moral Conduct) Scholarship (USTC) 2022
- Excellent President of Student Club (Origami Club, USTC) 2022
- First Prize, Chinese Mathematical Competition for University Students (Top 1%) 2022 & 2021
- Bronze Prize, Outstanding Freshmen Scholarship (Top 35%, USTC) 2020

TEACHING AND OUTREACH

- **Teaching Assistant: Linear Algebra B1** Mar. 2023 – Jul. 2023
University of Science and Technology of China (USTC), Hefei, China
 - Ranked **Top 3** among all teaching assistants at USTC in the university-wide TA evaluation.
 - Built the course homepage and served as the group leader, managing a course group with over 100 students.
 - Organized weekly problem-solving sessions during weekends, spent over 40 hours of personal teaching time, and engaged with more than 200 participants, showcasing strong teaching and communication skills.
 - Independently completed solution sets for over 100 post-course exercises and provided analysis and answers to all past exam papers.
- **Origami Club President** Jun 2021 – Jun 2022
University of Science and Technology of China
 - Designed and organized the first origami exhibition “Yue ran zhi shang” (Vividly Comes to Life on Paper) in USTC, responsible for venue decoration and maintaining order for two consecutive weeks.
 - Conducted origami teaching sessions twice per month, sharing the art of origami with over 200 participants.
 - Held special origami activities, such as Origami at Christmas and spray painting origami works.
- **Volunteer** Sep 2021 – Jan 2022
Hefei Chunyu Parent Support Center for Intellectually Disabled Children
 - Played basketball with autistic children and helped coaches to keep order in class once per week.

SELECTED COURSE PROJECTS

- **Airbnb Price Prediction in New York City** (CS 671 Theory & Alg Machine Learning, Duke University) – Achieved top 5 of 137 participants using advanced stacking strategies, combining XGBoost, LightGBM, and 20+ weak learners for robust generalization. [\[Report\]](#)[\[Code\]](#)

ADDITIONAL INFORMATION

Languages: English, Mandarin, Sichuanese dialect.

Programming & Software: Python (primary), C/C++, $\LaTeX$, Linux, HTML/CSS, MATLAB, R.

Interests: Origami and other visual arts (selected works available at [portfolio link](#)).